Incidence Geometry of Projected Operator Composition

Rank Protection, Carrier-Resolved Strata, and Promotion Limits

WuJun Chen
Independent Researcher
2026

Abstract

Problem. Let $A : \mathbb{C}^n \to \mathbb{C}^m$ and $B : \mathbb{C}^p \to \mathbb{C}^n$ be nonzero. Their product can nevertheless vanish. In a sectorized operator setting, this is the basic obstruction to promoting a Boolean support path through an intermediate sector to a nonzero projected composition.
Approach. We first study the closed free zero-product locus

$$Z = \{(A, B) : AB = 0\}$$

and its constructible nonzero-factor part. The identity $AB = 0 \iff \operatorname{im} B \subseteq \ker A$ turns product failure into an image–kernel incidence condition. We stratify this condition by $\operatorname{rank} A = r$ and by the double rank $(\operatorname{rank} A, \operatorname{rank} B) = (r, s)$. We then replace the free ambient by a declared common carrier decomposition and stratify by the carrier rank vectors $\mathbf{r} = (r_\beta)$ and $\mathbf{s} = (s_\beta)$.
Results. The fixed-r stratum has

$$\dim I_r = r(m + n - r) + p(n - r), \qquad \operatorname{codim} I_r = (m - r)(n - r) + pr.$$

The fixed-(r, s) incidence stratum has

$$\dim I_{r,s} = r(m + n - r) + s(n - r + p - s),$$

and has relative codimension rs inside the ambient rank-(r, s) matrix-pair stratum. In a common block structure, the zero-product locus is the product of the carrierwise loci. Its fixed-$\mathbf{r}$ codimension is

$$\sum_\beta [(m_\beta - r_\beta)(n_\beta - r_\beta) + p_\beta r_\beta],$$

and the fixed-$(\mathbf{r}, \mathbf{s})$ incidence condition has relative codimension $\sum_\beta r_\beta s_\beta$. The full diagonal pair ambient has zero-product codimension d, while a complementary-support represented family can lie entirely in incidence and therefore have pullback codimension zero. Full column rank of A, or full row rank of B, excludes nonzero zero-products. An exact carrier certificate upgrades the four canonical Rubik zero-route concentrations from numerical zeros to exact fixed-frame incidence. A separate 19-orbit restriction census remains a bounded Computational Observation.
Boundary. These results concern one routed product. They do not identify a support path with a routed product, a routed product with a full word, a full word with a commutator, or low-order commutator support with Lie depth. Likewise, neither free nor structured-ambient codimension determines the pullback incidence geometry of a representation-derived family. The structured theorem is fixed-frame: it does not compare endogenous sector frames or establish a universal incidence rate.

---

1 Introduction

The local question is elementary:

> Why can $A \neq 0$ and $B \neq 0$ coexist with $AB = 0$?

The answer is exact:

$$\operatorname{im} B \subseteq \ker A.$$

This condition is invisible to Boolean factor support. It depends on the relative position of two subspaces in the intermediate vector space. For projected operator blocks, it is therefore the first geometric datum that must be checked after a support path has been found.

The paper has four goals.

1. Give the rank-stratified geometry of the zero-product condition.
2. Refine that geometry for a declared common carrier decomposition.
3. Register a fixed-frame represented incidence profile with exact and numerical evidence kept separate.
4. Separate local incidence from every stronger promotion involving route sums, words, commutators, Lie depth, or representation-derived parameter families.

The analysis is static: the sector projectors, carrier projectors, and labelled operator family are fixed. Moving projectors, endogenous frame reconstruction, normal spectral charts, and general parameter-dependent pullbacks remain research directions.

The image–kernel obstruction is adjacent to two independent questions. First, Boolean graph paths need not survive projected matrix composition. Second, nonzero word terms need not survive antisymmetrization into a commutator. These neighboring failures motivate the promotion table below, but the corresponding theorems are not assumed here. The incidence and rank-protection criteria can serve as route-survival gates, but do not promote routed products to word or Lie accessibility.

Notation and Claim Layers

The notation is introduced below in the order required by the routed-product problem. The claim-status vocabulary is **Theorem** for exact statements under their stated hypotheses, **Computational Certificate** for a source-addressed finite exact or computational check explicitly assigned that status, **Computational Observation** for the declared Rubik, random-control, perturbation, and finite-atlas records, and **Research Program** for general represented pullback geometry and all stronger promotion statements that remain open. The complete status table appears in Section 9.

Let V be a finite-dimensional complex Hilbert space with a complete orthogonal sectorization

$$V = \bigoplus_{i \in \mathcal{I}} E_i, \qquad I = \sum_{i \in \mathcal{I}} Q_i, \qquad Q_i Q_j = \delta_{ij} Q_i.$$

Let $\mathcal{Y} = \{Y_a\}_{a \in \mathcal{A}}$ be a declared finite operator family. No commutant assumption is imposed. When a common carrier structure is invoked, it is separately declared as

$$V = \bigoplus_{\beta \in \mathcal{B}} H_\beta, \qquad I = \sum_{\beta \in \mathcal{B}} C_\beta,$$

with orthogonal carrier projectors C_β. The structured results require

$$[Q_i, C_\beta] = 0, \qquad [Y_a, C_\beta] = 0$$

for every registered i, a, β. These are additional hypotheses, not consequences of sectorization. They give carrier-resolved sector spaces

$$E_{i,\beta} = Q_i C_\beta V, \qquad E_i = \bigoplus_\beta E_{i,\beta}.$$

For a sector triple (i, k, j) and ordered operator pair (a, b), define

$$A^a_{ik} = Q_i Y_a Q_k : E_k \to E_i, \qquad B^b_{kj} = Q_k Y_b Q_j : E_j \to E_k.$$

Write

$$d_i = \dim E_i, \qquad d_k = \dim E_k, \qquad d_j = \dim E_j.$$

The routed projected composition is

$$T^{a,b}_{ikj} = Q_i Y_a Q_k Y_b Q_j = A^a_{ik} B^b_{kj}.$$

Our arrow convention is

$$j \longrightarrow k \longrightarrow i.$$

A support path records only

$$A^a_{ik} \neq 0, \qquad B^b_{kj} \neq 0.$$

It does not record whether $T^{a,b}_{ikj}$ is nonzero.
The full two-letter word block is a different object:

$$W^{a,b}_{ij} = Q_i Y_a Y_b Q_j = \sum_{k \in \mathcal{I}} T^{a,b}_{ikj}.$$

If the declared family is a skew-Hermitian Lie family $\mathcal{X} = \{X_g\}$, then the projected commutator is

$$C^{g,h}_{ij} = Q_i [X_g, X_h] Q_j = W^{g,h}_{ij} - W^{h,g}_{ij}.$$

These three objects – routed term, full word, and commutator – remain separate throughout the paper.

---

2 Image–Kernel Incidence

Theorem 2.1 (Image–Kernel Criterion)**.**
Let

$$A : \mathbb{C}^n \to \mathbb{C}^m, \qquad B : \mathbb{C}^p \to \mathbb{C}^n.$$

Then

$$AB = 0 \quad \Longleftrightarrow \quad \operatorname{im} B \subseteq \ker A.$$

Proof. *If $AB = 0$, then $A(Bv) = 0$ for every $v \in \mathbb{C}^p$, so every vector in $\operatorname{im} B$ lies in $\ker A$. Conversely, if $\operatorname{im} B \subseteq \ker A$, then $A(Bv) = 0$ for every v, hence $AB = 0$.*

2.1 Closed and constructible loci

Define

$$Z_{m,n,p} = \{(A, B) \in \mathrm{Mat}_{m\times n}(\mathbb{C}) \times \mathrm{Mat}_{n\times p}(\mathbb{C}) : AB = 0\}.$$

The entries of AB are polynomial in the entries of A and B, so $Z_{m,n,p}$ is a closed affine algebraic set.

The nonzero-factor incidence locus is

$$Z^{\times}_{m,n,p} = Z_{m,n,p} \cap \{A \neq 0\} \cap \{B \neq 0\}.$$

It is constructible, not generally a closed subvariety. Fixed exact-rank pieces are locally closed strata.

Proposition 2.2 (Ambient Generic Nonincidence)**.**
The complement of $Z_{m,n,p}$ is a nonempty Zariski-open dense subset of the free matrix-pair space. Consequently, $Z_{m,n,p}$ and $Z^{\times}_{m,n,p}$ have Lebesgue measure zero.

Proof. *The zero-product equations are polynomial, and they do not vanish identically on the ambient matrix-pair space. Thus their common zero set is a proper closed algebraic subset. Its complement is nonempty, Zariski open, and dense; every proper complex algebraic subset has real Lebesgue measure zero.*

This proposition concerns freely varying matrix pairs. It makes no statement about a constrained parameter family whose image may lie partly or entirely inside $Z_{m,n,p}$.

---

3 Rank-Stratified Geometry

Theorem 3.1 (Fixed-A-Rank Incidence Stratum)**.**
For $0 \leq r \leq \min(m, n)$, let

$$I_r = \{(A, B) : \operatorname{rank} A = r,\ \ AB = 0\}.$$

Then I_r is locally closed and

$$\dim I_r = r(m + n - r) + p(n - r).$$

Its codimension in $\mathrm{Mat}_{m\times n} \times \mathrm{Mat}_{n\times p}$ is

$$\operatorname{codim} I_r = (m - r)(n - r) + pr.$$

Proof. *The rank-r matrices A form a locally closed stratum of dimension $r(m + n - r)$. For fixed A, the condition $AB = 0$ requires every column of B to lie in the $(n - r)$-dimensional space $\ker A$. The fiber has dimension $p(n - r)$. Adding base and fiber dimensions gives the first formula. Subtracting from the ambient dimension $mn + np$ gives the codimension formula.*

For the nonzero-factor locus, the admissible A-ranks are exactly

$$1 \leq r \leq \min(m, n - 1).$$

The upper bound is $r < n$, not $r < \min(m, n)$. In particular, if $m < n$, a full-row-rank matrix with $r = m$ still has a nontrivial kernel and can participate in a nonzero zero-product.

Theorem 3.2 (Fixed Double-Rank Incidence Stratum).
Let

$$I_{r,s} = \{(A,B) : \operatorname{rank} A = r, \qquad \operatorname{rank} B = s, \qquad AB = 0\}.$$

This stratum is nonempty exactly when

$$0 \le r \le \min(m,n), \qquad 0 \le s \le \min(n-r,p).$$

When nonempty,

$$\dim I_{r,s} = r(m+n-r) + s(n-r+p-s).$$

Proof. *Choose a rank-r matrix A, contributing $r(m+n-r)$ dimensions. Its kernel has dimension $n-r$. A rank-s map $B : \mathbb{C}^p \to \ker A$ exists exactly when $s \le \min(n-r,p)$ and belongs to a rank-s matrix stratum of dimension $s(n-r+p-s)$.*

Corollary 3.3 (Relative Codimension by Rank Product).
Inside the ambient rank-pair stratum

$$\{(A,B) : \operatorname{rank} A = r,\ \operatorname{rank} B = s\},$$

the incidence condition $AB = 0$ has relative codimension

$$rs.$$

Proof. *The ambient rank-pair stratum has dimension*

$$r(m+n-r) + s(n+p-s).$$

Subtracting the dimension in Theorem 3.2 gives rs.

This formula isolates the alignment cost after the two ranks have already been fixed. Rank deficiency and image–kernel alignment are distinct conditions.

Corollary 3.4 (Square-Block Asymptotics).
For $m = n = p = d$ and $1 \le r \le d-1$,

$$\operatorname{codim} I_r = (d-r)^2 + dr = d^2 - dr + r^2.$$

The minimum is attained at the integer or integers nearest $d/2$, and equals

$$\left\lceil \frac{3d^2}{4} \right\rceil.$$

Thus the dominant nonzero-factor incidence stratum has codimension asymptotic to $3d^2/4$ in the $2d^2$-dimensional free matrix-pair space.

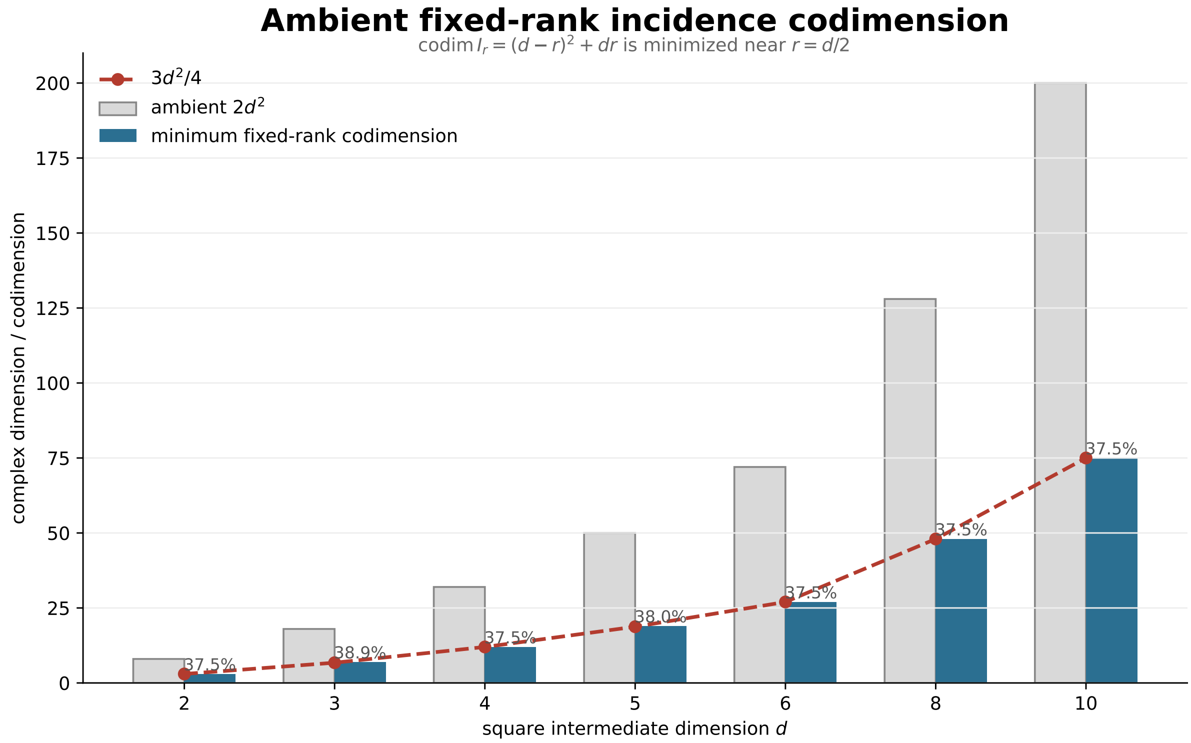

Figure 1. Exact fixed-rank codimension values for square blocks. The blue bars show the minimum of $\operatorname{codim} I_r = (d-r)^2 + dr$ over admissible nonzero ranks, against the $2d^2$ ambient dimension and the $3d^2/4$ asymptotic. This is an ambient free-matrix calculation, not a represented pullback codimension.

4 Rank Protection

Return to a sector triple:

$$A : E_k \to E_i, \qquad B : E_j \to E_k, \qquad d_k = \dim E_k.$$

Define

$$L = [\operatorname{rank} A = d_k], \qquad R = [\operatorname{rank} B = d_k].$$

The condition L means that A has full column rank, which is possible only when $d_i \geq d_k$. Similarly, the condition R means that B has full row rank, which is possible only when $d_j \geq d_k$.

Theorem 4.1 (Rank-Protected Product Survival)**.**
Let $A \neq 0$ and $B \neq 0$.

1. *If* $\operatorname{rank} A = d_k$*, then* $AB \neq 0$*.*
2. *If* $\operatorname{rank} B = d_k$*, then* $AB \neq 0$*.*

More quantitatively,

$$\|AB\|_F \geq \sigma_{\min}(A)\|B\|_F$$

when A has full column rank, and

$$\|AB\|_F \geq \sigma_{\min}(B)\|A\|_F$$

when B has full row rank.

Proof. Full column rank gives $\ker A = 0$, so Theorem 2.1 rules out $AB = 0$ for nonzero B. Applying the smallest-singular-value inequality to each column of B gives the first norm bound. For the dual statement, use $(AB)^* = B^* A^*$ and note that B^* has full column rank with the same positive singular values as B.

The condition

$$\operatorname{rank} A = \min(d_i, d_k)$$

is not sufficient for left protection. If $d_i < d_k$, such a matrix has maximum possible rank but still has a nonzero kernel. The analogous warning holds for B when $d_j < d_k$.

The five mutually exclusive audit classes are:

Class	Condition
both-protected	$L \wedge R$
left-only	$L \wedge \neg R$
right-only	$\neg L \wedge R$
unprotected-nonzero	$\neg L \wedge \neg R \wedge AB \neq 0$
unprotected-zero	$\neg L \wedge \neg R \wedge AB = 0$

Protected records are still required to evaluate AB numerically. Rank protection is used as a theorem-level cross-check, not as a branch that skips the product calculation.

5 Carrier-Structured Incidence

The free matrix-pair ambient treats every entry of A and B as an independent coordinate. A represented system often supplies a smaller ambient before any parameter pullback is considered. The first structured case is a declared common block decomposition of the source, intermediate, and target spaces.

Definition 5.1 (Common-Carrier Matrix-Pair Ambient)**.**
Let

$$E_i = \bigoplus_{\beta\in\mathcal{B}} E_{i,\beta}, \qquad E_k = \bigoplus_{\beta\in\mathcal{B}} E_{k,\beta}, \qquad E_j = \bigoplus_{\beta\in\mathcal{B}} E_{j,\beta},$$

where

$$m_\beta = \dim E_{i,\beta}, \qquad n_\beta = \dim E_{k,\beta}, \qquad p_\beta = \dim E_{j,\beta}.$$

The common-carrier matrix-pair ambient is

$$\mathcal{M}_{\mathcal{B}} = \left(\bigoplus_\beta \mathrm{Mat}_{m_\beta\times n_\beta}(\mathbb{C})\right) \times \left(\bigoplus_\beta \mathrm{Mat}_{n_\beta\times p_\beta}(\mathbb{C})\right).$$

Write $A = \bigoplus_\beta A_\beta$ and $B = \bigoplus_\beta B_\beta$. Its structured zero-product locus is

$$Z_{\mathcal{B}} = \{(A,B)\in\mathcal{M}_{\mathcal{B}} : AB = 0\}.$$

Zero-dimensional carrier blocks are permitted and contribute no coordinates.

Theorem 5.2 (Carrier Product Decomposition)**.**
The structured zero-product locus decomposes as

$$Z_{\mathcal{B}} = \prod_{\beta\in\mathcal{B}} Z_{m_\beta,n_\beta,p_\beta}.$$

Equivalently,

$$AB = 0 \quad\Longleftrightarrow\quad A_\beta B_\beta = 0 \text{ for every } \beta \quad\Longleftrightarrow\quad \mathrm{im}\, B_\beta \subseteq \ker A_\beta \text{ for every } \beta.$$

If the exact carrier supports

$$\mathcal{S}(A) = \{\beta : A_\beta \neq 0\}, \qquad \mathcal{S}(B) = \{\beta : B_\beta \neq 0\}$$

are disjoint, then $AB = 0$.

Proof. *Common-carrier reduction gives $AB = \bigoplus_\beta A_\beta B_\beta$. A direct-sum operator vanishes exactly when every block vanishes. Apply Theorem 2.1 to each block. The final statement follows because one factor of every carrier-local product is then zero.*

The disjoint-support clause is sufficient, not necessary. If the carrier supports overlap, zero product is still possible through carrier-local image–kernel alignment.

Theorem 5.3 (Fixed Carrier-Rank Stratum)**.**
Fix a carrier rank vector

$$\mathbf{r} = (r_\beta)_\beta, \qquad 0 \le r_\beta \le \min(m_\beta, n_\beta).$$

The locally closed stratum of $Z_{\mathcal{B}}$ with $\mathrm{rank}\, A_\beta = r_\beta$ for every β has dimension

$$\dim I^{\mathcal{B}}_{\mathbf{r}} = \sum_\beta \left[r_\beta(m_\beta + n_\beta - r_\beta) + p_\beta(n_\beta - r_\beta)\right]$$

and codimension in $\mathcal{M}_{\mathcal{B}}$

$$\operatorname{codim}_{\mathcal{M}_{\mathcal{B}}} I_{\mathbf{r}}^{\mathcal{B}} = \sum_{\beta} \left[(m_\beta - r_\beta)(n_\beta - r_\beta) + p_\beta r_\beta \right].$$

Proof. *By Theorem 5.2, the stratum is the product of the corresponding fixed-*r_β *strata. Apply Theorem 3.1 blockwise and add dimensions.*

Theorem 5.4 (Fixed Double Carrier-Rank Stratum)**.**
Fix also $\mathbf{s} = (s_\beta)_\beta$*. The structured fixed-double-rank incidence stratum is nonempty exactly when*

$$0 \le s_\beta \le \min(n_\beta - r_\beta, p_\beta)$$

for every β*. When nonempty, it has dimension*

$$\dim I_{\mathbf{r},\mathbf{s}}^{\mathcal{B}} = \sum_{\beta} \left[r_\beta(m_\beta + n_\beta - r_\beta) + s_\beta(n_\beta - r_\beta + p_\beta - s_\beta) \right].$$

Inside the structured rank-$(\mathbf{r}, \mathbf{s})$ *matrix-pair stratum, its relative codimension is*

$$\sum_{\beta} r_\beta s_\beta.$$

Proof. *Apply Theorem 3.2 and Corollary 3.3 independently on every carrier block, then take the product.*

Corollary 5.5 (Diagonal Ambient and Pullback Boundary)**.**
For the full diagonal pair ambient with d *scalar carrier blocks,*

$$A = \operatorname{diag}(a_1, \dots, a_d), \qquad B = \operatorname{diag}(b_1, \dots, b_d),$$

one has

$$AB = 0 \quad \Longleftrightarrow \quad a_\beta b_\beta = 0 \text{ for every } \beta.$$

The zero-product locus is a union of coordinate linear spaces of dimension d *in the* $2d$*-dimensional diagonal pair ambient. Its codimension is therefore* d*, not one. Its nonzero-factor part is nonempty for* $d \ge 2$*, for example by choosing disjoint nonempty supports for* A *and* B*.*
By contrast, the represented map

$$\Phi : \mathbb{C}^2 \longrightarrow \mathcal{M}_{\mathcal{B}}, \qquad \Phi(u, v) = (\operatorname{diag}(u, 0), \operatorname{diag}(0, v))$$

satisfies $AB = 0$ *identically. Hence* $\Phi^{-1}(Z_{\mathcal{B}}) = \mathbb{C}^2$ *has pullback codimension zero, while both factors are nonzero on the open set* $uv \ne 0$*.*
This gives three different questions:

$$\operatorname{codim}_{\mathit{free}} Z, \qquad \operatorname{codim}_{\mathit{structured}} Z_{\mathcal{B}}, \qquad \operatorname{codim}_{\Theta} \Phi^{-1}(Z_{\mathcal{B}}).$$

None may be substituted for another.

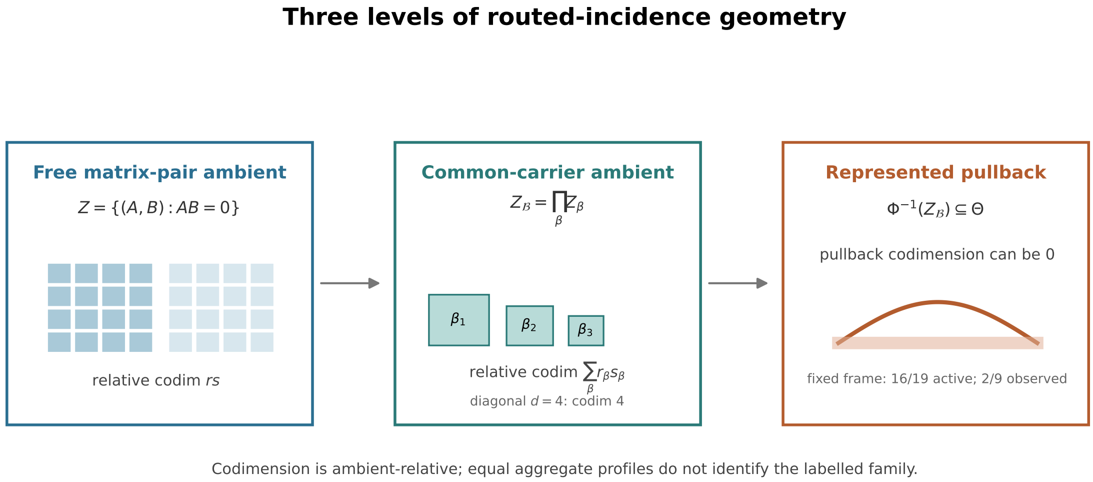

Figure 2. Free matrix-pair incidence supplies a baseline, a declared common carrier decomposition replaces it by a product of blockwise incidence loci, and a represented parameter family may meet or lie inside that structured locus. Codimension is ambient-relative at every stage.

6 Fixed-Frame Routed-Incidence Profiles

The carrier theorems become represented statements only after a fixed sector frame, labelled operator family, and exact or numerical policy are supplied. For the route (i, k, j, a, b), common-carrier reduction gives

$$A^a_{ik} = \bigoplus_{\beta} A^a_{ik,\beta}, \qquad B^b_{kj} = \bigoplus_{\beta} B^b_{kj,\beta}.$$

Let $\Omega_{Q,y}$ be the finite set of declared route tuples for which both factors are supported. A **fixed-frame routed-incidence profile** records:

1. the identities of $\{Q_i\}$, $\{C_\beta\}$, and the labelled alphabet $\mathcal{Y}$;
2. the route scope, threshold and rank policies, and evidence level;
3. carrier rank signatures and the five rank-protection classes;
4. exact carrier-forced zeros, carrier-local image–kernel zeros, and unresolved numerical zeros as distinct mechanisms;
5. both $N_0/|\Omega_{Q,y}|$ and $N_0/N_{\text{unprotected}}$.

The two denominators answer different questions. An unavailable denominator is not replaced by zero. An exact mechanism label requires exact carrier reduction and exact support masks; a floating-point threshold supplies only a numerical diagnostic.

Proposition 6.1 (Transported-Frame Covariance)**.**
Suppose a unitary U transports two represented data sets by

$$Q'_i = UQ_iU^{-1}, \qquad C'_\beta = UC_\beta U^{-1}, \qquad Y'_{\alpha(a)} = UY_aU^{-1},$$

for explicit label bijections. Then corresponding routed products are unitarily conjugate. Their Frobenius norms, carrier rank vectors, rank protection classes, carrier-incidence mechanisms, and aggregate profile counts agree under the transported policy.

__Proof.__ Substitute the transported projectors and operators into each routed product. Adjacent $U^{-1}U$ factors cancel, leaving the unitary conjugate of the original route. Unitary conjugation preserves ranks and Frobenius norms, and the transported carrier projectors preserve the carrier labels up to the declared bijection.

The converse is false: equality of aggregate profiles does not identify the labelled generator family or the frame. Profile equality is therefore a coarse represented invariant, not a reconstruction theorem.

---

7 Promotion Limits

The local incidence theorem resolves only the first matrix-composition gate. For fixed i, k, j, g, h, the following implications are invalid without additional hypotheses:

Local datum	Stronger conclusion	Missing promotion condition
$A \neq 0$, $B \neq 0$	$T^{g,h}_{ikj} \neq 0$	image–kernel nonalignment or rank protection
some $T^{g,h}_{ikj} \neq 0$	$W^{g,h}_{ij} \neq 0$	no cancellation in the sum over k
$W^{g,h}_{ij} \neq 0$ and $W^{h,g}_{ij} \neq 0$	$C^{g,h}_{ij} \neq 0$	no antisymmetric cancellation
low-order Lie support	finite or exact $D_{\mathrm{Lie}}(i, j)$	higher Hall data and closure/saturation certificate
high free or structured codimension	rare represented incidence	nondegenerate pullback or transversality of the parameter map

Equivalently,

$$\begin{aligned} \text{support path} &\not\Longrightarrow T^{g,h}_{ikj} \neq 0, \\ T^{g,h}_{ikj} \neq 0 &\not\Longrightarrow W^{g,h}_{ij} \neq 0, \\ W^{g,h}_{ij}, W^{h,g}_{ij} \neq 0 &\not\Longrightarrow C^{g,h}_{ij} \neq 0. \end{aligned}$$

The first failure is controlled exactly by image–kernel incidence for the declared route. The second is a sum-over-routes cancellation problem. The third is an antisymmetrization problem. None is interchangeable with the others.

7.1 Free, structured, and represented incidence

Let a constrained family be parameterized by

$$\Phi : \Theta \longrightarrow \mathcal{M}_{\mathcal{B}}.$$

Its incidence locus is

$$\Phi^{-1}(Z_{\mathcal{B}}).$$

Theorem 5.3 determines the structured-ambient codimension, but not the dimension of this pullback. A promotion from ambient genericity to a represented family requires, at minimum, evidence that $\Phi(\Theta)$ is not contained in $Z_{\mathcal{B}}$ and a suitable transversality or nondegeneracy statement on the relevant carrier-rank strata. Symmetry or prescribed support may force a family into incidence despite high free or structured codimension, as Corollary 5.5 shows.

---

8 Exact and Computational Case Studies

The displayed integer examples are exact theorem-level examples. The canonical carrier and rotation-orbit records are source-addressed Computational Certificates. The routed censuses, random controls, perturbations, and finite Lie atlas are Computational Observations. None is used to prove the abstract incidence theorems.

8.1 Declared numerical policy

The projected-composition audit uses:

Quantity	Policy
support	$\|A\|_F, \|B\|_F > 10^{-8}$
numerical rank	singular values above $\max(10^{-12}, 10^{-9}\sigma_{\max})$
product nonzero	$\|AB\|_F > 10^{-12} + 10^{-10}\|A\|_F\|B\|_F$
normalized product	$\eta(A,B) = \|AB\|_F/(\|A\|_F\|B\|_F)$
image–kernel action	$\|AU_B\|_F$, where U_B spans $\operatorname{im} B$
subspace residual	$\delta_{\mathrm{inc}}(A,B) = \|(I - P_{\ker A})U_B\|_F$

Here the columns of U_B are an orthonormal basis of $\operatorname{im} B$ and $P_{\ker A}$ is the orthogonal projector onto $\ker A$. The reported δ_{inc} is not divided by $\sqrt{\operatorname{rank} B}$. Small $\eta(A,B)$ is a numerical alignment diagnostic, not an exact incidence proof. Every protected record is still multiplied, and a numerically protected zero-product triggers an assertion.

8.2 Exact integer examples

Five explicit integer matrix pairs realize all five audit classes. In the table, $r_A = \operatorname{rank} A$ and $r_B = \operatorname{rank} B$.

Example	(m,n,p)	(r_A, r_B)	Outcome
both	$(2,2,2)$	$(2,2)$	$AB \neq 0$
left only	$(3,2,1)$	$(2,1)$	$AB \neq 0$
right only	$(1,2,3)$	$(1,2)$	$AB \neq 0$
unprotected, nonzero	$(2,2,2)$	$(1,1)$	$AB \neq 0$
unprotected, zero	$(2,2,2)$	$(1,1)$	$AB = 0$

A concrete integer realization is

1. both protected: $A = I_2$ and $B = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$;
2. left only: $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$;
3. right only: $A = \begin{pmatrix} 1 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$;
4. unprotected and nonzero: $A = B = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$;
5. unprotected and zero: $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$.

Their ranks and products can be checked directly. In particular, the last two examples demonstrate that a lack of protection is necessary but not sufficient for incidence.

8.3 Exact carrier certificate and fixed-frame restriction census

The structured Rubik registration uses the four declared physical carriers

$$V = V_{\mathrm{cp}} \oplus V_{\mathrm{ep}} \oplus V_{\mathrm{co}} \oplus V_{\mathrm{eo}}.$$

An exact finite certificate over $\mathbb{Z}[\zeta_3]$, with $\zeta_3^2 + \zeta_3 + 1 = 0$, constructs the nine canonical joint quarter-turn/half-turn projectors by Lagrange interpolation. Writing each projector as $P_\ell = N_\ell / d_\ell$, it verifies entrywise

$$N_\ell^2 = d_\ell N_\ell, \qquad N_\ell N_m = 0 \quad (\ell \neq m),$$

and, for a common denominator D,

$$\sum_\ell \frac{D}{d_\ell} N_\ell = DI.$$

The exact traces give the nine dimensions and sum to 228. Thus idempotence, pairwise orthogonality, completeness, and dimension are checked independently. For distinct physical carriers $\beta \neq \gamma$, the certificate also checks

$$C_\beta \rho(g) C_\gamma = C_\beta \rho(g)^* C_\gamma = C_\beta(\rho(g) - \rho(g)^*) C_\gamma = 0$$

for all 18 labelled generators. Hence the represented operators, their adjoints, and the anti-Hermitian numerators all preserve the same exact carrier decomposition.

The coefficient arrays use `int64`, but exactness is not inferred from the absence of a runtime warning. Before each identity construction, addition, subtraction, integer scaling, adjoint, matrix multiplication, and trace, the validator records a conservative absolute coefficient bound. All 387 audited operations pass; the largest bound is $121031884800 < 2^{63} - 1$. Trace accumulation is performed with Python integers. These prior bounds close the fixed-width arithmetic obligation without replacing the 228-dimensional matrix products by object-array arithmetic.

For the four zero-based sector triples

$$(2, 6, 8), \qquad (5, 6, 8), \qquad (8, 6, 2), \qquad (8, 6, 5),$$

the endpoint carrier masks are disjoint. Theorem 5.2 therefore proves $A^a_{ik} B^b_{kj} = 0$ exactly for every registered operative pair on these triples. This certificate upgrades the zero status of the numerator recorded by the numerical Rubik audit below. It does not certify that every other supported product is exactly nonzero.

A separate exact combinatorial enumeration considers axis-balanced, inverse-closed subsets of the 18 labelled Rubik generators. On each of the three axes, one independently retains none of its turns, its four quarter turns, its two half turns, or both classes. The three four-way choices give $4^3 = 64$ labelled families; removing the empty family leaves 63. Quotienting these 63 labelled families by the 24 orientation-preserving cube rotations gives 19 orbit representatives. Every family is evaluated against the same canonical nine-sector frame; no sector decomposition is recomputed. Among the 19 representative rows, 16 have supported anti-Hermitian routes and three consist only of operators that vanish under this registration. Every active row has the finite numerical rate

$$\frac{N_{\text{carrier-forced zero}}}{N_{\text{supported}}} = \frac{2}{9}.$$

Across the 16 unweighted active representatives this is $4224/19008 = 2/9$. Every zero in the numerator has an exact carrier-disjointness witness, but the complementary nonzero set is still thresholded numerical evidence. The ratio is therefore a bounded Computational Observation for the declared family index, not an exact or universal law.

As a non-identifiability control, let

$$\mathcal{Y}_0 = \mathcal{Y}_{18} \setminus \{L^2, R^2\}, \qquad \mathcal{Y}_2 = \mathcal{Y}_{18} \setminus \{F^2, B^2\}.$$

These are distinct labelled 16-generator families. An exact combinatorial certificate places them in the same orientation-preserving rotation orbit. Proposition 6.1 gives profile equality only after representation equivariance and transport of the sector and carrier frames are supplied. Thus equal sector counts, sector dimensions, or aggregate incidence profiles do not identify which labelled half-turn pair was removed.

8.4 Rubik routed-product registration

The computational Rubik case study fixes:

- the nine declared center-decomposition sectors, whose dimensions are

$$(20, 2, 39, 26, 1, 39, 66, 8, 27);$$

- the 18 skew-Hermitian operators $X_g = (\rho(g) - \rho(g)^*)/2$;
- all ordered operator pairs, including repeated operators;
- all sector triples with off-diagonal factor legs $i \neq k$ and $k \neq j$.

For each order-two half turn, unitarity gives $\rho(g)^* = \rho(g)^{-1} = \rho(g)$, and therefore $X_g = 0$. The audit confirms six zero operators, at zero-based indices $2, 5, 8, 11, 14, 17$, with maximum norm exactly zero in the declared realization. Consequently, all nonzero-factor routed records arise from the twelve quarter-turn operators. This explains the count $144 = 12^2$ for each exceptional sector triple, even though the declared family contains 18 operators and the enumeration initially considers all ordered pairs.

This anti-Hermitian-part registration is not the numerical matrix-logarithm family used in Paper V's S_4 case study [2]. In particular, a half turn vanishes here but generally has a nonzero logarithmic generator. The two papers' R_1^{Lie} and R_2^{Lie} censuses therefore cannot be compared without an explicit observable-family alignment.

The corrected census is:

Class	Count
both-protected	0
left-only	0
right-only	0
unprotected-nonzero	2016
unprotected-zero	576
total nonzero-factor routes	2592

The 576 machine-zero products are concentrated in four zero-based sector triples:

(i, k, j)	Count
$(2, 6, 8)$	144
$(5, 6, 8)$	144
$(8, 6, 2)$	144
$(8, 6, 5)$	144

Across these records,

$$\max \|AB\|_F = 2.467 \times 10^{-16}, \qquad \max \eta(A, B) = 6.737 \times 10^{-17}.$$

The minimum factor norm is 1.803. The maximum image–kernel action residual is 1.210×10^{-15} and the maximum subspace distance residual is 9.485×10^{-15}. Taken alone, these are machine-zero image–kernel alignment records in the declared complex128 realization. The independent carrier certificate above establishes the exact zero status of these 576 routes under the exact canonical projector and operator registration.

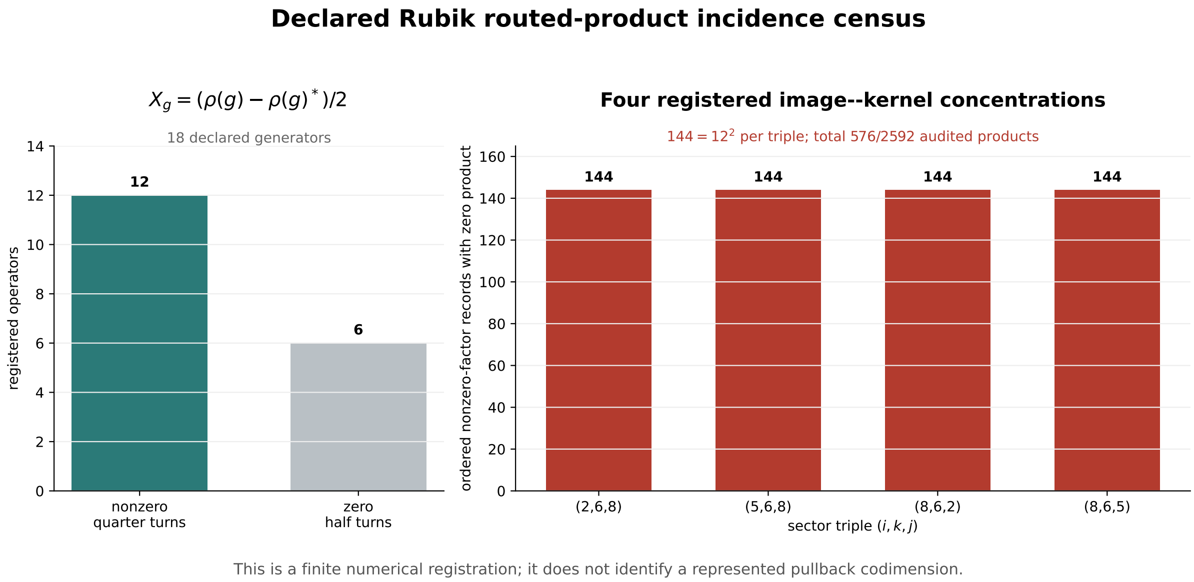

Figure 3. The declared Rubik routed-product census. Six half-turn operators vanish under the anti-Hermitian-part registration, leaving twelve nonzero quarter-turn operators and therefore $144 = 12^2$ nonzero-factor records in each of four registered incidence concentrations. The complex128 census supplies the route counts and residuals; the separate exact carrier certificate proves the zero status of the four concentrations.

8.5 Random and rank-deficient controls

Five dense random systems contribute 1620 routed records. Every record is both-protected and has nonzero product.

Five controls with an 80% rank-deficient block-generation probability contribute another 1620 records:

Class	Count
both-protected	122
left-only	274
right-only	274
unprotected-nonzero	950
unprotected-zero	0

Thus, rank deficiency creates unprotected routes but does not, by itself, force image–kernel incidence.

For independent dense square matrix pairs, the normalized product audit gives:

d	$\min\eta$	$q_{0.001}$	$q_{0.01}$	median
2	0.1068	0.2322	0.3461	0.7070
3	0.2235	0.3017	0.3678	0.5744
4	0.2460	0.3144	0.3610	0.4975
5	0.2444	0.3122	0.3463	0.4455

Each dimension uses 100000 samples. No sample falls below relative thresholds 10^{-8}, 10^{-10}, or 10^{-12}. This is an implementation and threshold sanity check consistent with Proposition 2.2; it is not evidence for operator or Lie completion.

8.6 Perturbation and finite full-array atlas

Constructed incidence pairs in dimensions $2, 3, 4$ were perturbed at $\varepsilon = 10^{-6}, 10^{-4}, 10^{-2}$. All 900/900 perturbed products became nonzero at the declared threshold.

A separate finite Lie audit registers 80 eight-dimensional skew-Hermitian systems in four mask families. It stores complete generator-indexed R_1^{Lie} arrays, complete commutator-indexed R_2^{Lie} arrays, complete D_{Lie} matrices, each per-depth support array, each cumulative support array, and deterministic SHA-256 hashes.

Each of the four threshold-defined low-order array classes contains 20 systems. Within every class:

- the complete D_{Lie} arrays agree elementwise;
- all per-depth and cumulative support arrays agree elementwise;
- the results are stable at thresholds $10^{-9}, 10^{-8}, 10^{-7}$;
- the cumulative generator–commutator basis passes the closure audit described below.

At rank tolerance 10^{-8}, the cumulative dimensions by mask family, listed through the first empty round, are

1. family 0: $(3, 6, 14, 28, 49, 63, 63)$, first empty round 6;
2. family 1: $(3, 5, 10, 20, 40, 59, 63, 63)$, first empty round 7;
3. family 2: $(3, 5, 10, 18, 31, 43, 51, 58, 61, 63, 63)$, first empty round 10;
4. family 3: $(3, 6, 14, 32, 63, 63)$, first empty round 5.

An empty round means that every commutator of a basis element in the newest layer with every declared generator produces no new vector after projection against the cumulative basis. The closure audit additionally checks all generator–basis pairs, not solely those involving the newest layer. If $\mathcal{L} = \operatorname{span}\{L_a\}$ is the final numerical basis, it forms

$$R_{\mathrm{cl}} = \big[(I - P_{\mathcal{L}})\,\mathrm{vec}[X_g, L_a]\big]_{g,a}.$$

Across all 80 systems, the final dimension is 63, the smallest retained singular value of the vectorized basis matrix is at least 0.999999999998, the largest singular value of R_{cl} is at most 7.175×10^{-13}, and the largest individual closure residual is 6.627×10^{-13}. Thus all generator–basis commutators lie in the final span to well below the declared rank tolerance. This is a numerical closure certificate for the finite atlas, not an exact-arithmetic Lie-algebra theorem.

No disagreement occurs in this finite atlas. This is an observation about the declared four families, not a theorem that $(R_1^{\mathrm{Lie}}, R_2^{\mathrm{Lie}})$ determines D_{Lie}.

9 Claim Status and Boundary

The table uses the four claim levels. Corollaries and exact finite examples belong to the Theorem level; refuted or unclaimed promotions are listed separately as boundaries.

Claim	Status
Image–Kernel Criterion	Theorem
$Z = \{AB = 0\}$ is closed; $Z^\times$ is constructible	Theorem
fixed-r dimension and codimension	Theorem
fixed-(r, s) dimension and relative codimension rs	Theorem
square-block dominant codimension	Theorem
left/right rank protection and norm bounds	Theorem
common-carrier product decomposition and carrierwise image–kernel criterion	Theorem
fixed-$\mathbf{r}$ and fixed-$(\mathbf{r}, \mathbf{s})$ structured dimensions	Theorem
structured relative codimension $\sum_\beta r_\beta s_\beta$	Theorem
diagonal-ambient codimension d and complementary-support pullback example	Theorem
transported-frame covariance of routed-incidence profiles	Theorem
existence of the five displayed exact integer examples	Theorem
63 labelled axis-balanced families and 19 rotation orbits	Computational Certificate
exact canonical carrier masks and exact zero status of the four Rubik concentrations	Computational Certificate
fixed-frame 19-orbit restriction census and active-row 2/9 rate	Computational Observation
six half-turn zeros and corrected Rubik routed census	Computational Observation
random quantiles and perturbation breakdown	Computational Observation
full-array finite Lie atlas agreement and closure residual	Computational Observation
low-order Lie support determines Lie depth	Research Program
general represented pullback incidence and transversality	Research Program

The boundary is explicit: graph-to-route, route-to-word, and word-to-commutator promotions are not claimed. Furthermore, free or carrier-structured ambient codimension does not determine represented pullback geometry without additional hypotheses. Fixed-frame profile equality does not identify a labelled operator family.

There is no theorem or conjecture in this paper asserting

$$(R_1^{\mathrm{Lie}}, R_2^{\mathrm{Lie}}) \implies D_{\mathrm{Lie}}.$$

---

10 Research Program

10.1 Conditional Low-Order Promotion Problem

A future promotion theorem would need explicit, independently checkable hypotheses rather than an undefined richness condition. Candidate gates include:

Gate	Candidate requirement
H1 low-order coverage	every finite-depth target already appears at the declared R_1^{Lie} or R_2^{Lie} layer
H2 route survival	every required routed product avoids image–kernel incidence
H3 cancellation control	route sums and antisymmetrizations do not erase the required channels, or are repaired in a declared Hall layer
H4 closure	the numerical or exact Lie closure is certified saturated

H1 is an exclusionary hypothesis, not an observed general principle. Indeed, it fails in the declared S4 numerical realization of Paper V: two registered channels have $R_1^{\mathrm{Lie}} = R_2^{\mathrm{Lie}} = 0$ but first appear at $D_{\mathrm{Lie}} = 2$ [2]. The table is a research checklist, not a theorem statement.

10.2 Pullback incidence

Theorem 5.3 resolves one declared common-carrier ambient. Let a more constrained represented family be given by

$$\Phi : \Theta \longrightarrow \mathcal{M}_{\mathcal{B}}.$$

Determine:

1. whether $\Phi(\Theta)$ is contained in an incidence stratum;
2. whether Φ is transverse to the relevant carrier-rank strata;
3. how symmetry changes the pullback codimension;
4. whether ranks remain constant on the declared parameter chart.

The object of interest is the pullback locus $\Phi^{-1}(Z_{\mathcal{B}})$, not the free or structured ambient codimension alone. Corollary 5.5 gives an identically incident control, not a general pullback-dimension theorem.

10.3 Structural stability and moving charts

The reported perturbation audit varies free matrix entries. A structural stability theorem would instead perturb inside a representation-preserving or sector-preserving family. Moving projectors require normal spectral charts and coherent continuation before incidence can be compared across parameters.

10.4 Hub recurrence

The four Rubik zero-product triples share the intermediate sector $k = 6$. Whether such incidence concentration relates to transport centrality remains open. A formal relation would require aligned operator families and a theorem connecting graph, routed, and representation-derived data.

---

11 Conclusion

Projected factors compose precisely when the image of the right factor is not contained in the kernel of the left factor. Thus

$$AB = 0 \quad \Longleftrightarrow \quad \operatorname{im} B \subseteq \ker A$$

is the local geometric obstruction, while full column rank of A or full row rank of B protects a nonzero product. On a fixed double-rank stratum, the incidence condition has relative codimension rs, cleanly separating rank deficiency from the additional image–kernel alignment cost.

When a common carrier decomposition is part of the data, the zero-product locus is the product of its carrierwise incidence loci. Fixed carrier-rank vectors replace rs by $\sum_\beta r_\beta s_\beta$. This structured codimension is still not a represented pullback codimension: the full diagonal ambient has codimension d, while a complementary-support family can lie inside incidence with pullback codimension zero.

In the declared Rubik realization, the finite complex128 audit records 2016 unprotected nonzero products and 576 zero products among 2592 nonzero-factor routes. The route counts and residuals are Computational Observations. An independent exact carrier certificate proves the zero status of the four registered concentrations. Across the fixed canonical frame, the 19-orbit restriction census retains 2/9 only as a bounded numerical pattern because the complementary nonzero products have not all been certified exactly.

These conclusions stop at routed matrix composition. They do not identify a support path with a routed product, a routed product with a full word, a word with a commutator, or low-order support with finite Lie depth. Rank protection, carrier structure, represented pullback geometry, and promotion limits are therefore distinct parts of one static composition problem.

---

12 Related Work and Novelty Boundary

The matrix inequalities and singular-value bounds are standard [7]. Determinantal and incidence strata are standard objects in algebraic geometry [6, 5, 4]. The carrier-structured locus is a direct product of these standard incidence strata; the dimension formulas follow by blockwise addition. The new role here is to keep that structured ambient distinct from both the free ambient and a representation-derived pullback. The distinction between Boolean zero patterns and evaluated matrix products is adjacent to qualitative matrix theory and structural controllability [1, 8]. Routed products may also be viewed through path and quiver evaluation, although this paper does not construct a path-algebra quotient [9].

Related work separates support graphs from projected composition and direct support from words, commutators, and low-depth Lie data [3, 2]. Paper III's block-local obstruction covers all finite projected compositions under its physical-block hypotheses; the carrier-disjoint clause of Theorem 5.2 is its one-product specialization. The rank-vector incidence geometry and fixed-frame profile developed here do not re-own that all-length theorem.

Appendix A Computational Artifacts

The following repository artifacts support the exact incidence tables and the computational case studies. The default directory is `experiments/paper7/`; paths are relative to that directory.

Artifact	Role	Short path
A1	exact fixed-rank and fixed-double-rank tables	`validation/incidence_variety_codim.py`
A2	routed-product, rank-protection, and incidence audit	`validation/rank_protected_bridge_audit.py`
A3	full-array typed low-order and Lie-depth audit	`validation/atlas_r2_boundary.py`
A4	generated incidence tables	`results/incidence_geometry.json`, `.txt`
A5	generated projected-composition witnesses	`results/projected_composition_audit.json`, `.txt`
A6	generated complete-array Lie atlas	`results/full_array_lie_atlas.json`, `.txt`
A7	carrier-structured formula and boundary validator	`validation/structured_incidence_geometry.py`
A8	generated structured-incidence record	`results/`
A9	fixed-frame profile promotion validator	`validation/register_fixed_frame_profiles.py`
A10	source-addressed fixed-frame profile registration	`results/`

From the repository root, run an executable artifact as `python experiments/paper7/<short path>`. A4 contains the fixed-rank tables; A5 records thresholds, exact examples, the Rubik operator-family declaration, zero-product witnesses, random controls, and perturbations; A6 records full arrays, per-layer and cumulative supports, dimensions, closure certificates, and hashes. A8 evaluates the structured formulas, diagonal correction, and codimension-zero represented control exactly. A10 binds the selected fixed-frame source artifacts and their SHA-256 digests. Its source package is the Paper VII exploratory `incidence_profiles/` directory; the registration excludes every endogenous-frame and cross-frame artifact. A8 and A10 each serialize JSON and text under the paper-local `results/` directory.

Every zero-product witness records

$$(d_i, d_k, d_j), \quad (\operatorname{rank} A, \operatorname{rank} B), \quad \sigma^+_{\min}(A), \quad \sigma^+_{\min}(B),$$

together with $\|A\|_F$, $\|B\|_F$, $\|AB\|_F$, $\eta(A, B)$, $\|AU_B\|_F$, the image-to-kernel distance, sector indices, and operator indices.

The array hashes detect artifact drift. Mathematical equality claims in the finite atlas use numpy.array_equal on the complete arrays rather than hash equality alone.

All listed artifacts are available in the RIME repository.

References

[1] Richard A. Brualdi and Herbert J. Ryser. *Combinatorial Matrix Theory*, volume 39 of *Encyclopedia of Mathematics and its Applications*. Cambridge University Press, Cambridge, 1991.

[2] WuJun Chen. Boolean support does not determine commutator accessibility: Direct support, exact cancellation, and low-depth hall bridges. *Preprint*, 2026. Paper V of the RIME program, version 2.0. DOI: `https://doi.org/10.5281/zenodo.21634007`.

[3] WuJun Chen. Support-graph reachability and matrix-composition obstructions: Image–kernel mismatch with a Rubik-cube case study. *Preprint*, 2026. Paper III of the RIME program, version 2.0. DOI: `https://doi.org/10.5281/zenodo.21583070`.

[4] David Eisenbud. *Commutative Algebra: With a View Toward Algebraic Geometry*, volume 150 of *Graduate Texts in Mathematics*. Springer, New York, 1995.

[5] William Fulton. *Intersection Theory*. Springer, Berlin, 2 edition, 1998.

[6] Joe Harris. *Algebraic Geometry: A First Course*, volume 133 of *Graduate Texts in Mathematics*. Springer, New York, 1992.

[7] Roger A. Horn and Charles R. Johnson. *Matrix Analysis*. Cambridge University Press, Cambridge, 2 edition, 2013.

[8] Ching-Tai Lin. Structural controllability. *IEEE Transactions on Automatic Control*, 19(3):201–208, 1974.

[9] Ralf Schiffler. *Quiver Representations*. CMS Books in Mathematics. Springer, Cham, 2014.